MTMed3D: A Multi-Task Transformer-Based Model for 3D Medical Imaging

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***Abstract*—In the field of medical imaging, AI-assisted techniques such as object detection, segmentation, and classification are widely employed to alleviate the workload of clinicians. However, most approaches rely on single-task models, overlooking shared information across tasks and leading to redundant computation and increased deployment complexity. In this work, we propose MTMed3D, an end-to-end multi-task Transformer-based framework for 3D brain tumor analysis that jointly performs detection, segmentation, and classification. MTMed3D uses a Transformer as the shared encoder to generate multi-scale features, followed by CNN-based task-specific decoders. We evaluate MTMed3D on BraTS 2018 and 2019, achieving competitive performance across tasks, with particularly strong results in detection. Compared with separately trained single-task models, MTMed3D improves detection performance while reducing computational cost and achieving a 2.3× inference speedup. To the best of our knowledge, this is the first Transformer-based multi-task framework that jointly addresses detection, segmentation, and classification for 3D brain tumor analysis, presenting its potential as an efficient framework for comprehensive tumor analysis in clinical settings. The code is available at https://github.com/fanlimua/MTMed3D.git.**

***Index Terms*—3D Medical Imaging, Multi-task Model, Segmentation, Classification, Detection**

I. Introduction

In the real world, medical diagnoses often require multi-perspective assessment of tumor morphology, making detection, segmentation, and classification crucial in medical imaging. With advances in deep learning, significant performance gains have been achieved in medical image analysis. However, most medical image analysis models are single-task models, which are effective for individual tasks but have limitations compared with multi-task models. They typically learn from one type of label, preventing them from using knowledge from related tasks. This limitation is critical in medical imaging, where annotated data are scarce and costly, leading to less effective training. In contrast, multi-task models share representations across tasks, enabling complementary feature learning and mitigating data scarcity. Moreover, single-task models incur a higher computational cost by requiring separate training and inference for each task. Instead, multi-task models share parameters across tasks, enabling a single model to perform multiple tasks with reduced cost and inference time.

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CNN-based methods are widely adopted in medical imaging [1], [2], but their limited kernel size restricts modeling long-range dependencies. Meanwhile, transformer-based models [3], [4] have emerged as a promising alternative for medical image analysis. Transformers model long-range dependencies with self-attention, but suffer from quadratic complexity [5]. Swin Transformers [6] address this by applying attention within local windows and using shifted windows for cross-window interaction, enabling efficient local-global modeling.

Given these challenges, efficient multi-task learning models have become increasingly important for practical medical imaging applications. Building on Swin Transformer advancements, we propose an end-to-end Transformer-based multi-task model for 3D brain tumor analysis, where a shared encoder jointly performs segmentation, detection, and classification, reducing redundancy and improving efficiency. In summary, our main contributions are as follows:

- We propose MTMed3D, to our knowledge, the first end-to-end multi-task Transformer-based model for 3D brain tumor segmentation, detection, and classification.
- We evaluate MTMed3D on the Multimodal Brain Tumor Segmentation Challenge (BraTS) [7] 2018 and 2019, achieving overall promising performance, with strong detection results over existing methods on BraTS 2018.
- We analyze the impact of shared feature representations and show that MTMed3D improves tumor detection over single-task models, while reducing computational cost and achieving a 2.3× inference speedup. We further validate our design through comprehensive ablation studies.

II. Related Work

Multi-task Learning. Multi-task learning (MTL) jointly learns multiple related tasks, leveraging shared knowledge to improve performance. Park et al. [8] proposed a ViT-based model for joint lung classification, detection, and segmentation. They adopt federated and split learning, training a shared transformer centrally with task-specific components on separate machines and datasets. In contrast, MTMed3D is trained end-to-end on a single machine using the same input for all tasks. Moreover, their ViT encoder has 66.37M parameters, significantly larger than our 8.06M Swin Transformer encoder. Amyar et al. [9] present an automatic classification and segmentation tool for helping to diagnose COVID-19 pneumonia using chest CT imaging. Yan et al. [10] introduced

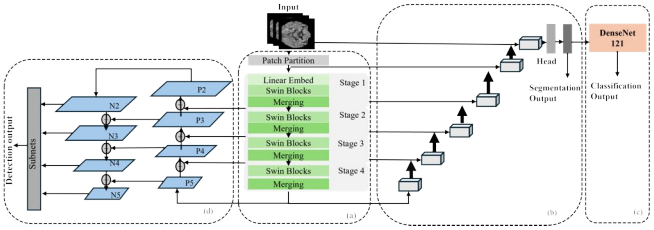


Fig. 1. Overview of MTMed3D: (a) Swin Transformer-based encoder, (b) Segmentation Decoder, (c) Classification Decoder, (d) Detection Decoder.

a multi-task universal lesion analysis network designed for the joint detection, tagging, and segmentation of lesions across various body parts. This network is built upon an improved Mask R-CNN framework featuring three head branches and a 3D feature fusion strategy.

Transformer-based applications. Transformer-based models have gained increasing attention in medical image analysis. For segmentation, TransUNet [3] was the first Transformer-based framework for medical image segmentation, integrating a Transformer layer into the bottleneck of a U-Net for multi-organ segmentation. Swin UNETR [4] employs a Swin Transformer encoder with a CNN decoder for feature extraction and reconstruction. In classification tasks, Lu et al. [11] proposed a two-stage framework that first employs contrastive pre-training to extract meaningful representations and then aggregates features using a Transformer-based sparse attention module for label prediction. Zhang et al. [12] introduced a two-stage Transformer-based framework for COVID-19 diagnosis, using UNet for lung segmentation followed by Swin Transformer feature extraction and aggregation for 3D classification. For object detection tasks, Shen et al. [13] proposed a Convolutional Transformer (COTR) network for end-to-end polyp detection. COTR consists of a CNN for feature extraction, Transformer encoder layers interleaved with convolutional layers for feature encoding and recalibration, Transformer decoder layers for object querying, and a feed-forward network for detection prediction. This architecture effectively leverages the strengths of both CNNs and Transformers, enabling COTR to outperform DETR on two different datasets.

III. MTMed3D Model Design

MTMed3D is a multi-task Transformer-based model for 3D brain tumor analysis that consists of a shared encoder and three task-specific decoders (Figure 1). In designing MTMed3D, we address the following challenges. ***Challenge 1:*** Enabling efficient parameter sharing without high computational overhead. **Solution:** We adopt hard parameter sharing, leveraging a shared encoder to improve efficiency. ***Challenge 2:*** Preventing task imbalance during joint training due to gradient dominance. **Solution:** We employ GradNorm [14] to adaptively balance task losses. ***Challenge 3:*** BraTS datasets lack detection annotations. **Solution:** We generate detection labels from segmentation annotations.

A. Encoder

MTMed3D uses a shared Swin Transformer encoder with four hierarchical stages, as shown in Figure 1(a), each comprising two Swin blocks followed by a patch merging layer. A 3D medical image is partitioned into patches, embedded into a latent space, and passed through four stages to produce hierarchical feature representations. A Swin Transformer block, following [6], consists of window-based and shifted window multi-head self-attention with Layer Normalization, residual connections, and MLP layers. After each stage, patch merging reduces the feature-map resolution, enabling the encoder to generate multi-scale representations. This shared encoder addresses Challenge 1 by enabling parameter sharing across tasks and reducing redundancy.

B. Decoders

Detection Decoder. For detection, we adopt a modified RetinaNet [15] with a Swin Transformer encoder as the backbone and replace FPN [24] with PANet [25]. The encoder produces multi-scale features from stages $i = 2, 3, 4$ and the bottleneck ($i = 5$), which are unified in channel dimension and progressively upsampled and fused via skip connections to generate feature maps P_i. PANet further enhances feature propagation by introducing a bottom-up pathway (N_2–N_5), where each N_i is downsampled, concatenated with P_{i+1}, and enhanced to produce N_{i+1}. A box regression subnet with convolutional layers, GroupNorm, and ReLU predicts object coordinates.

Segmentation Decoder. U-Net has shown strong performance in medical image segmentation, inspiring many U-shaped designs [4], [26]. We adopt a Swin UNETR [4]-based decoder for segmentation. The Swin Transformer encoder extracts multi-scale features, which are fed to the decoder via skip connections. Features from each encoder stage are processed with residual blocks, and the decoder progressively upsamples from the bottleneck using deconvolution layers, concatenating skip features to enhance predictions. A final convolution layer produces the segmentation logits.

Classification Decoder. For classification, following [27], we use segmentation outputs as inputs and place the classification branch after segmentation, adopting DenseNet-121 [28] for efficient feature reuse and improved gradient flow.

C. Optimization

MTMed3D employs three decoders with a weighted sum of losses: $L_{\text{total}} = w_1 L_{\text{seg}} + w_2 L_{\text{cls}} + w_3 L_{\text{det}}$, where L_{seg}, L_{cls}, and L_{det} denote Dice, Focal, and SmoothL1 losses, respectively, and w_1, w_2, and w_3 are their corresponding weights. To address Challenge 2, we employ GradNorm [14] to balance gradients by optimizing the multi-task loss. The multi-task loss is defined as $L(t) = \sum_{i=1}^{T} w_i(t) L_i(t)$, where the sum is over T tasks at step t. GradNorm normalizes gradient magnitudes across tasks by computing the L_2 norm $G_w^{(i)}(t)$ of the weighted loss $w_i(t)L_i(t)$, and the mean gradient $\overline{G}_w(t)$ across tasks: $\overline{G}_w(t) = \mathbb{E}_{\text{task}}[G_w^{(i)}(t)]$. The inverse training rate is defined as $\widetilde{L_i}(t) = L_i(t)/L_i(0)$, and $r_i(t) = \widetilde{L_i}(t)/\mathbb{E}_{\text{task}}[\widetilde{L_i}(t)]$, where $r_i(t)$ reflects the relative training speed of task i. Tasks

TABLE I
RESULTS ON BRATS 2018 AND 2019. WT: WHOLE TUMOR, TC: TUMOR CORE, ET: ENHANCING TUMOR.

Task	**Segmentation**						**Classification**			**Detection**			
Metric	DSC (WT	TC	ET)	HD (WT	TC	ET)	Acc	Sen	Spe	mAP@0.1:0.5	mAP@0.5	mAR@0.1:0.5	mAR@0.5
BraTS 2018	0.9082	0.8183	0.8038	5.4346	6.9773	5.6794	0.9298	0.9286	0.9333	0.9711	0.8650	0.9844	0.9298
BraTS 2019	0.8814	0.8084	0.7879	13.1085	9.9521	7.7802	0.9701	0.9804	0.9375	0.8820	0.7368	0.9187	0.8209

TABLE II
DETECTION PERFORMANCE OF DIFFERENT METHODS.

Methods	**mAP@0.1:0.5**	**mAP@0.5**	**mAR@0.1:0.5**	**mAR@0.5**
RetinaNet [15]	0.8664	0.6768	0.8850	0.7193
nnDetection [16]	–	0.8360	–	–
MedYOLO [16]	–	0.8610	–	–
MTMed3D	**0.9711**	**0.8650**	**0.9844**	**0.9298**

TABLE III
SEGMENTATION PERFORMANCE COMPARISON WITH SOTA METHODS.

Methods	**DSC**				**HD**			
	WT	**TC**	**ET**	**Avg.**	**WT**	**TC**	**ET**	**Avg.**
AutoReg [17]	0.8839	0.8154	0.7664	0.8219	5.9044	**4.8091**	3.7731	4.8289
MIC-DKFZ [18]	0.8781	0.8062	0.7788	0.8210	6.0300	5.0800	**2.9000**	**4.6700**
DeepSCAN [19]	0.8859	0.7992	0.7318	0.8056	5.5185	5.5347	3.4808	4.8447
MultiDeep [20]	0.8842	0.7960	0.7775	0.8192	5.4681	6.8773	2.9366	5.0940
MTMed3D	**0.9082**	**0.8183**	**0.8038**	**0.8434**	**5.4346**	6.9773	5.6794	6.0304

with slower loss reduction have larger $r_i(t)$ and are assigned higher target gradients. GradNorm updates task weights by minimizing the L_1 loss between actual and target gradients: $L_{\text{grad}}(i) = \left| G_w^{(i)}(t) - \overline{G}_w(t) \cdot r_i(t) \right|$. The weights w_i are optimized via backpropagation, dynamically balancing tasks.

IV. EXPERIMENT

A. Setting

Dataset. We evaluate MTMed3D on BraTS 2018 and 2019 [7]. To address Challenge 3, we generate detection labels by deriving bounding boxes from segmentation annotations.

Implementation Details. Experiments are conducted on an NVIDIA RTX 4070 Ti GPU. Images are normalized and cropped to 96^3 for memory limits. Learning rates are set to $1\text{e}{-4}$ (segmentation) and $1\text{e}{-5}$ (detection/classification), using AdamW with cosine annealing. Since the BraTS server only supports segmentation, we use 5-fold cross-validation on the training set for consistent multi-task evaluation and report the fold with the highest overall performance as an upper-bound reference for contextual comparison.

Evaluation Metrics. We evaluate segmentation with Dice Similarity Coefficient (DSC) and Hausdorff Distance (HD), detection with mean Average Precision (mAP) and mean Average Recall (mAR), and classification with Accuracy (Acc), Sensitivity (Sen), and Specificity (Spe). mAP/mAR@0.1:0.5 are computed over IoU thresholds of 0.1-0.5 with a 0.05 step, and @0.5 uses IoU=0.5. Efficiency is measured by Floating Point Operations (FLOPs), Parameters (Params), and latency. Overall performance is reported using *Task Avg.*, defined as the average of task-specific metrics after normalization to [0,1].

TABLE IV
CLASSIFICATION PERFORMANCE COMPARISON WITH SOTA METHODS.

Methods	**Accuracy**	**Sensitivity**	**Specificity**
Mask R-CNN (2D) [21]	0.9630	0.9350	0.9720
ConvNet (3D) [21]	0.9710	0.9470	0.9680
CNN (2D) [22]	0.9715	0.9818	**0.9855**
SE-ResNeXt (2D) [23]	**0.9745**	**0.9835**	0.9493
MTMed3D	0.9298	0.9286	0.9333

B. Results

1) Evaluation on BraTS 2018 and 2019 Datasets: Table I reports results across tasks on two datasets. For segmentation, the model achieves high DSC on both datasets, particularly on the WT region. For classification, the model performs strongly on both datasets, with higher accuracy and sensitivity on BraTS 2019. Detection performance varies across datasets, due to distribution differences affecting shared representations. Despite this, the model maintains competitive results across tasks, demonstrating consistent multi-task performance.

2) Comparison with State-of-the-Art Methods:

Detection Results. As BraTS mainly targets segmentation and classification, we implement a 3D RetinaNet (ResNet-FPN) baseline on BraTS 2018 for comparison. We also compare with nnDetection and MedYOLO [16] at mAP@0.5 ("-" indicates unavailable metrics). Table II shows that MTMed3D outperforms all baselines, indicating strong detection performance.

Segmentation Results. Table III compares our segmentation results with SOTA methods on BraTS 2018. MTMed3D achieves the highest DSC across all regions. It also achieves the best HD on WT, but is weaker on TC and ET. Despite this, the model achieves competitive segmentation performance. **Insight 1:** The compared methods are single-task and heavily rely on ensembling for performance, whereas MTMed3D achieves competitive results with a single end-to-end model, demonstrating its efficiency.

Classification Results. Table IV compares classification performance on BraTS 2018. MTMed3D does not outperform existing methods, as prior approaches typically use 2D slices or task-specific enhancements. Instead, MTMed3D performs end-to-end 3D multi-task learning without dedicated pipelines or pre-trained weights, demonstrating the feasibility of a unified multi-task framework. **Insight 2:** MTMed3D performs well on segmentation and detection, but classification remains challenging as it requires both global and local features. Swin-based design may suppress sparse local cues, limiting class separability and suggesting that hybrid architectures, as explored in ResMT [29], may benefit glioma grading.

TABLE V
PERFORMANCE COMPARISON OF MULTI-TASK AND SINGLE-TASK MODELS AND ABLATION STUDY ON MULTI-TASK COMPONENTS.

Task	Segmentation								Classification			Detection		Task Avg.
Metric	DSC (WT	TC	ET	Avg.)	HD (WT	TC	ET	Avg.)	Acc	Sen	Spe	mAP@0.1:0.5	mAR@0.1:0.5	
Single-task	0.8936	**0.8203**	0.7453	0.8235	5.8720	**4.8691**	**4.6084**	**5.2418**	0.9298	0.9487	0.8888	0.8182	0.8596	0.8920
w/ FPN	**0.9152**	0.7927	0.7807	0.8295	7.2079	9.5378	6.0401	7.5953	**0.9649**	**1.0000**	0.8667	0.9341	0.9630	0.9164
w/o GradNorm	0.8418	0.7053	0.7457	0.7643	23.2645	16.6405	8.3818	16.0956	0.8070	**1.0000**	0.2667	0.3589	0.4834	0.7200
w/ Deeper Encoder	0.9044	0.7971	0.7170	0.8062	9.3141	8.8320	7.9112	8.6858	0.9123	0.9286	0.8667	0.9684	0.9825	0.9019
MTMed3D	0.9082	0.8183	**0.8038**	**0.8434**	**5.4346**	6.9773	5.6794	6.0304	0.9298	0.9286	**0.9333**	**0.9711**	**0.9844**	**0.9245**

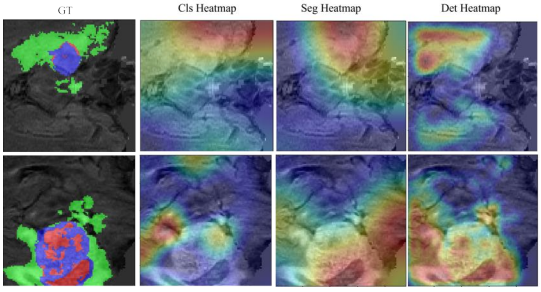


Fig. 2. Grad-CAM visualizations of MTMed3D across tasks.

3) Qualitative Results: To analyze task interactions, we visualize Grad-CAM [30] heatmaps for all tasks. As shown in Figure 2, all tasks focus on tumor regions, indicating shared representations. Classification shows both global and local attention patterns, while segmentation and detection emphasize tumor and candidate regions, respectively. This highlights task-specific preferences and competition in multi-task learning.

4) Efficiency Analysis: We compare the efficiency of MTMed3D with three independently trained single-task models using the same architecture, as shown in Table VI (Bottom). MTMed3D reduces FLOPs by 46.64% and parameters by 47.80%. Additionally, MTMed3D achieves a 2.3× inference speedup and provides a more computationally and memory-efficient solution than separate single-task models. We also analyze the computational cost of model components. Table VI (Top) shows that segmentation is more computationally expensive than detection, making detection a better choice in resource-constrained settings.

C. Ablation Studies

We conduct ablation studies on BraTS 2018 (Table V).

Multi-task vs Single-task Models. Compared to single-task models, MTMed3D achieves higher DSC for segmentation, comparable classification performance, and better detection, resulting in superior overall performance with improved computational efficiency. **Insight 3:** The improvement of detection can be attributed to shared representations that better capture features hard to learn in isolation, suggesting multi-task learning benefits tasks with limited information.

Deeper Encoder. We compare our [2,2,2,2] configuration (Swin blocks per stage) with deeper variants [2,2,6,2] and [2,2,18,2] [6]. The deeper [2,2,6,2] (*w/ Deeper Encoder*) model shows lower performance and efficiency, while [2,2,18,2] fails due to memory limits. This indicates that [2,2,2,2] delivers strong performance with high efficiency.

TABLE VI
EFFICIENCY ANALYSIS OF MTMED3D. TOP: COMPONENT-WISE EFFICIENCY; BOTTOM: MODEL-LEVEL EFFICIENCY.

Model / Component	FLOPs (G) ↓	Params (M) ↓	Latency (s) ↓
Swin Encoder	14.269	8.063	0.041
Seg Decoder	318.146	54.129	0.051
Cls Decoder	23.164	11.289	0.006
Det Decoder	40.971	3.218	0.007
Single-task Models	743.23	146.95	0.23
MTMed3D	**396.55**	**76.70**	**0.10**

Detection Branch. We compare PANet with an FPN-based detection branch. The variant (*w/ FPN*) shows lower segmentation and detection performance, and despite slightly higher classification accuracy, has a worse *Task Avg.* This demonstrates the advantage of PANet, whose additional pathway enhances feature propagation and improves overall performance.

Without Task Balancing Optimizer. We also study the effect of removing GradNorm. Without it (*w/o GradNorm*), multi-task optimization becomes unstable, and overall performance degrades. With GradNorm, balanced gradients lead to more consistent multi-task performance.

V. DISCUSSION

Existing datasets like MSD [31] and MedMNIST [32] support cross-organ multi-task learning, while our MTMed3D focuses on a single organ, enabling richer disease insights. Due to the lack of suitable 3D brain tumor datasets, we extend BraTS by generating detection labels from segmentation masks. Despite this, ablation study (*w/o GradNorm*) shows that segmentation-dominated training does not benefit detection, suggesting detection performance is not a proxy for segmentation. Grad-CAM further reveals distinct attention patterns, suggesting that different tasks focus on complementary visual cues. Beyond architecture, MTMed3D achieves strong detection performance and competitive segmentation results, while classification remains challenging. Grad-CAM suggests that 3D glioma grading requires both global context and local details, highlighting challenges in shared representations. These findings provide practical insights for future work on multi-task feature sharing and global–local trade-offs.

VI. CONCLUSION

We present MTMed3D, an end-to-end multi-task model for 3D brain tumor analysis. We evaluate it on the BraTS

datasets, demonstrating promising and well-balanced performance across tasks. Compared with SOTA single-task models, MTMed3D shows advantages in detection and segmentation. In particular, detection benefits from multi-task training, achieving higher performance and consistent gains over single-task counterparts. MTMed3D is also more efficient, requiring fewer resources and enabling faster inference than single-task alternatives while maintaining strong performance. Ablation studies validate our design choices. Overall, our results demonstrate the potential of Swin Transformer–based shared encoders for efficient multi-task learning in medical imaging.

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